

CIS

Cognitive Intelligence System

USER MANUAL

cisnew.vercel.app

Version 1.0 · 2026

Table of Contents

Table of Contents.....	2
1. Introduction	4
2. Core Concepts	4
3. Getting Started.....	5
Starting a New Investigation	5
4. The Dashboard	5
Top Bar	5
Pool Status Row.....	5
Dashboard Panels.....	5
5. Signal Intake	6
Steps to Submit a Signal	6
Live SI Preview (Right Panel)	6
After Submission	6
6. The Signal Pool.....	6
Lifecycle Filter Tabs.....	6
Filter Bar.....	7
Signal Row	7
Signal Detail Drawer.....	7
7. The Hypothesis Board.....	7
Hypothesis Types.....	7
Hypothesis Card.....	8
Adding Evidence	8
Hypothesis Resolution.....	8
8. Contradictions	8
Resolving a Contradiction.....	8
Resolution Steps	8
9. Cognitive Briefing.....	9
Briefing Contents.....	9
Exporting	9
10. Domain Manager.....	9
Independence Matrix	9
11. Analytics	10
Headline Metrics.....	10
Charts.....	10

12. Signal Status Colour Reference	10
13. What CIS Does Not Do	11
14. Glossary.....	11

1. Introduction

CIS (Cognitive Intelligence System) is a structured investigation platform for analysts, investigators, safety reviewers, fraud examiners, intelligence analysts, and medical case consultants. It manages observations through a defined lifecycle, detects structural patterns across independent information domains, generates and evaluates hypotheses about shared causes, and produces auditable cognitive briefings.

CIS is designed for investigations where:

- The explanation requires synthesising information from independent sources
- Premature closure is a documented failure risk
- The investigation must be auditable after the fact
- Missing a pattern across individually sub-threshold signals is a known cause of bad outcomes

2. Core Concepts

Signal: A structured observation. Any piece of information registered into the investigation. Not just anomalies, a signal can be a reading within normal parameters that is structurally incongruent with its context.

Signal Pool: The managed set of all signals in the investigation. Signals move through states:

Candidate → Admitted → Retained → Connected / Quarantined / Elevated → Assessed → Resolved → Archived

Domain: An independent organisational or technical source. Domains are the structural units for cross-domain analysis.

SI Score (Structural Incongruence Score): A composite score measuring how much a signal's structural properties deviate from what its context predicts. Four dimensions: rate of change, direction, cross-dimension relationship, constraint configuration. Minimum SI of 0.25 is required for Signal Pool admission.

Loss Points (LP-1 through LP-7): Seven specific points where signals are systematically lost in standard investigation:

- LP-1: Threshold exclusion
- LP-2: Single-period discard
- LP-3: Domain boundary loss
- LP-4: Explanation displacement
- LP-5: Aggregation failure
- LP-6: Significance decay
- LP-7: Closure-induced discard

CIS protects against all seven.

HCL Hypothesis: A structural claim that signals from independent domains share a common structural cause. Generated when cross-domain signal correspondence exceeds the independence threshold.

Cognitive Briefing: Structured output produced at any point: active signal pool summary, highest-significance retained signals, active hypotheses with plausibility, quarantined contradictions, resolved findings, and loss points flagged.

3. Getting Started

Starting a New Investigation

1. Navigate to `cisnew.vercel.app`
2. **Case Title:** Enter a descriptive title in the large input field
3. **Access Code:** Enter your access code in the field below (used to return to your case)
4. The system creates your case and takes you to the Dashboard

Important: No account required. Cases are anonymous. Keep your access code, it is the only way to return to your case.

4. The Dashboard

Route: `/cases/:id`

The Dashboard is the primary overview of your investigation.

Top Bar

- **Case Title:** displayed prominently
- **Access Code:** click to copy to clipboard
- **GENERATE BRIEFING:** produces a new Cognitive Briefing
- **Export:** export case data
- **Settings:** case configuration

Pool Status Row

Displays signal counts by status, each in its status colour. Click any status badge to jump to the Signal Pool filtered by that status.

`Candidate` · `Admitted` · `Retained` · `Connected` · `Quarantined` · `Elevated` · `Assessed` · `Resolved` · `Archived` · `Expired`

Dashboard Panels

- **Active Hypotheses (top left):** Top 3 active hypotheses. Shows title, type badge, plausibility bar (0–1.0), and evidence count. Click to open the Hypothesis Board.
- **Quarantine Alert (top right):** Count of quarantined contradictions. Lists the first 3 with description previews. Red/amber treatment. Click RESOLVE to address them.

- **High-Significance Signals (middle left):** Top 5 retained signals by significance score: signal preview, SI score, significance score, and domain tag.
 - **LP Flags (middle right):** Loss points flagged in this session. Shows LP code, signal reference, and brief description.
 - **Open Questions (bottom left):** Signals with $SI \geq 0.5$ and no assigned hypothesis. Structurally important observations not yet explained.
 - **Last Briefing (bottom right):** Time of last briefing and summary preview. GENERATE NEW button.
-

5. Signal Intake

Route: `/cases/:id/intake`

Signal Intake is the primary workflow page where you register new observations.

Steps to Submit a Signal

5. **Select Domain:** Choose from the dropdown. Use + ADD DOMAIN to register a new one.
6. **Set Observation Period:** Enter a number or leave as “auto (current period)”
7. **Describe the Observation:** Be precise, include what you measured, what the reading was, and what structural context this observation exists within.
8. Click SUBMIT SIGNAL

Live SI Preview (Right Panel)

As you type, the right panel shows:

- Preliminary SI score breakdown across 4 dimensions
- Expected admission decision
- Similar active signals already in the pool (AI-assisted)

After Submission

- Full signal card with all scores
- Admission decision (Admitted / sub-threshold / rejected with reason)
- Any cross-domain connections detected
- Any hypotheses triggered

Click SUBMIT ANOTHER to register the next observation.

6. The Signal Pool

Route: `/cases/:id/signals`

Lifecycle Filter Tabs

Filter Bar

- **Domain:** filter by source domain
- **Min SI:** minimum SI score threshold
- **Min Significance:** minimum significance score
- **Sort:** by significance (descending) or created (descending)

Signal Row

Each signal displays:

- Status dot (coloured by status)
- Domain tag
- Signal content (truncated, expandable)
- SI score bar (0–1.0)
- Significance score
- Observation period
- Action buttons: ELEVATE | QUARANTINE | RESOLVE | VIEW DETAIL

Signal Detail Drawer

Click VIEW DETAIL to open the full signal view (slides in from right):

- Full content
- SI scoring: 4 dimension bars + composite score
- NL4-B significance scoring: 6 criterion bars + composite
- Lifecycle event log (all state transitions with timestamps and reasons)
- Connected signals (if any)
- Contradictions (if any)
- Evidence references (if this signal is cited in hypothesis evidence)

7. The Hypothesis Board

Route: `/cases/:id/hypotheses`

The Hypothesis Board manages all hypotheses about what is causing the patterns in your investigation.

Hypothesis Types

Type	Description
HCL	Hidden Common Link: signals from independent domains share a structural cause
SI_CLUSTER	Multiple SI-bearing signals from a single domain
PATTERN	Temporal accumulation warranting aggregate assessment
INVESTIGATOR	Manually created by the investigator

Hypothesis Card

Each card shows:

- Title and type badge
- **Plausibility bar (0–1.0)**: the key number
- Generation source (SHG auto-generated vs investigator)
- Connected signals count
- Evidence count (supporting / contradicting / contextual)
- Last updated timestamp
- EXPAND | ADD EVIDENCE | RESOLVE buttons

Adding Evidence

Click ADD EVIDENCE to open the modal:

9. Select evidence type: SUPPORTING | CONTRADICTING | CONTEXTUAL
10. Set weight (0.0–1.0)
11. Enter content
12. Optionally link to a signal
13. Preview the expected plausibility delta before confirming

Hypothesis Resolution

- **CONFIRMED**: Plausibility ≥ 0.85 with no active contradicting evidence
- **FALSIFIED**: Plausibility ≤ 0.10 with strong falsifying evidence

8. Contradictions

Route: `/cases/:id/contradictions`

The Contradiction Ledger tracks all cases where two signals about the same structural state contradict each other. Quarantined signals cannot be resolved or expired while in Quarantine.

Resolving a Contradiction

Resolution requires one of three bases:

RC-1: Calibration Invalidation: The source of one signal is confirmed unreliable. Document which source and why.

RC-2: Falsification: One signal predicts an observable condition that is demonstrably absent. Document the predicted condition and the evidence of absence.

RC-3: Domain Resolution: Someone with direct observational access confirms which condition is present. Document who confirmed and how.

Resolution Steps

14. Click RESOLVE on a quarantined row
 15. Review both signals in full
 16. Select resolution type (RC-1, RC-2, or RC-3)
 17. Enter resolution basis
 18. Indicate which signal is accepted (if applicable)
 19. Click CONFIRM RESOLUTION
-

9. Cognitive Briefing

Route: `/cases/:id/briefing`

The Cognitive Briefing is the structured epistemic summary of your investigation at a point in time. Briefings are timestamped and stored, the investigation's epistemic state is auditable at every point.

Briefing Contents

- Pool Status: signal counts with colour-coded status indicators
- AI Summary: narrative paragraph summarising the investigation state
- Active Signals: highest-significance signals with scores
- Active Hypotheses: all active hypotheses with plausibility
- Quarantine: all unresolved contradictions
- Loss Points Flagged: any LP-1 through LP-7 events detected
- Open Questions: high-SI signals with no assigned hypothesis
- Resolved This Session: what was resolved in this session

Exporting

- **PDF:** via print-to-PDF
 - **Plain text:** for archival or handoff
-

10. Domain Manager

Route: `/cases/:id/domains`

Domains are the independent organisational or technical sources in your investigation. Proper domain management is essential for cross-domain analysis.

Independence Matrix

A grid showing all domain pairs. Each cell must be explicitly set:

- **INDEPENDENT:** signals from these domains can be compared for cross-domain analysis
 - **DEPENDENT:** these domains share a source; cross-domain analysis is not valid
 - **UNSET:** the relationship is not defined; cross-domain analysis is disabled for this pair
-

Important: CIS cannot run cross-domain analysis for any domain pair unless their relationship is explicitly set to INDEPENDENT.

11. Analytics

Route: /analytics

The public Analytics Dashboard shows aggregate investigation quality data across all CIS investigations (anonymous, no case content).

Headline Metrics

- Total investigations run
- Signal admission rate (% of candidates admitted)
- HCL hypothesis confirmation rate
- Assumption closure rate

Charts

- **Loss Point Distribution:** Bar chart of LP-1 through LP-7 counts. Shows where signals are systematically lost.
- **Signal Lifecycle Funnel:** Submitted → Admitted → Retained → Connected → Assessed → Resolved. Drop-off visible at each stage.
- **Hypothesis Type Breakdown:** HCL vs SI_CLUSTER vs PATTERN vs Investigator. Confirmation rate by type.

12. Signal Status Colour Reference

Status	Meaning
CANDIDATE	Submitted observation, pending admission scoring
ADMITTED	Passed SI and significance thresholds, in the pool
RETAINED	Preserved per WSP (sub-threshold but structurally significant)
CONNECTED	Cross-domain correspondence detected, flagged for SHG
QUARANTINED	In contradiction with another signal, frozen pending resolution
ELEVATED	Flagged for priority attention
ASSESSED	Hypothesis coverage assigned
RESOLVED	Explanation confirmed, lifecycle complete
ARCHIVED	Closed, retained in record
EXPIRED	Minimum retention elapsed, removed from active pool

13. What CIS Does Not Do

CIS is a decision-support tool. It surfaces patterns and generates hypotheses. It does not:

- Reach conclusions: investigators assess all outputs
- Replace domain expertise: SI scoring, plausibility assessment, and resolution all require human judgment
- Store personal data: cases are anonymous, no accounts required
- Issue authoritative continuity state determinations: all outputs are advisory estimates
- Assign governance consequences: CIS outputs inform decisions, they do not make them
- Make cross-system causal attribution: causal attribution requires domain expert review
- Override domain expert assessment: if expert assessment conflicts with CIS output, the conflict is flagged for human governance review

14. Glossary

Term	Definition
AP-v1	The seven protective constructs: CR (Contradiction Resolution), WSP (Weak Signal Preservation), CDA (Cross-Domain Analysis), SI (Structural Incongruence), HCL (Hidden Common Link), LD (Loss Defense), FA (Forensic Auditability)
Cognitive Briefing	Structured epistemic summary of an investigation at a point in time
Domain	An independent organisational or technical source of signals
HCL	Hidden Common Link: a structural hypothesis that signals from independent domains share a common cause
LP-1 through LP-7	The seven loss points where signals are systematically lost in standard investigation
NL4-B	The six-criterion significance scoring framework
Plausibility	Hypothesis confidence score (0-1.0). ≥ 0.85 = confirmed; ≤ 0.10 = falsified
SHG	Structural Hypothesis Generation: detects cross-domain correspondence and generates HCL hypotheses
SI Score	Structural Incongruence Score: how much a signal deviates from what its context predicts. Minimum 0.25 for pool admission.
SLS	Signal Lifecycle System: manages signals from observation through resolution
Signal Pool	The managed set of all signals in the investigation

Term	Definition
WSP	Weak Signal Preservation: sub-threshold signals are aggregated rather than discarded individually